Indian Institute of Technology, Bhilai

CSP203: Software Tools & Technologies Laboratory

Lab Activity - 6

Full Marks: 45 *Deadline: 05:30 PM, 21-08-2025*

Submission Instructions:

1) Answer all the questions.
2) Write all the scripts, execute and take screenshots of the scripts, name the scripts files as <Your_Roll_No>_<Lab_Activity_No>_<Question_No>.
3) Prepare a report pdf which contains the question number and screenshot of script code and corresponding outputs. Report quality will contain **5 marks**. You should prepare a zip folder which will contain the report and all the script files.
4) Naming of the zip file should strictly follow the given format:
   <Roll_No>_<Name>_<Activity_No>
   If your name is Shareen, roll number is B24CS055 then the filename should be:
   B24CS055_Shareen_04.pdf
5) **Plagiarism is strictly prohibited.**

Objectives:

1) Understand how to define and use awk in Bash scripts to modularize code and perform calculations on CSV data.
2) Apply array handling and parameter passing in Bash to process multiple files and compute required values.
3) Practice using Linux commands such as awk, grep, and wc for efficient text and file data processing.

4) Enhance problem-solving and scripting skills by working with real-world scenarios like student grading, word counting, and email filtering.

TASK 1.

You are given a CSV file named students.csv. This file contains student records in the following format: (RollNo,Name,Maths,Science,English)

Each row contains a student's roll number, name, and their marks in three subjects.

Tasks:

1. Read the CSV file using the awk command.
2. Skip the header (first) line.
3. For each student:
    a. Calculate the total marks.
    b. Calculate the average marks, rounded to two decimal places.
4. Based on the average, assign a grade as follows:
    "Distinction" if average ≥ 75
    "Pass" if 40 ≤ average < 75
    "Fail" if average < 40
5. Print the result for each student in the following format:
    Roll No: <roll>, Name: <name>, Total: <total>, Average: <average>, Grade: <grade>
6. Save the output to a file named report_awk.txt.

Constraints:

Use only the awk command for processing the file, calculating marks, and printing output.

Do not use cut, sed, or Bash arithmetic in your solution.

Sample Input (students.csv):

```
RollNo,Name,Maths,Science,English
101,John,85,78,92
102,Alice,45,50,42
103,Bob,70,68,72
104,Eve,90,88,95
105,Tom,30,35,28
```

Expected Output (report_awk.txt):

Roll No: 101, Name: John, Total: 255, Average: 85.00, Grade: Distinction
Roll No: 102, Name: Alice, Total: 137, Average: 45.67, Grade: Pass
Roll No: 103, Name: Bob, Total: 210, Average: 70.00, Grade: Pass

Roll No: 104, Name: Eve, Total: 273, Average: 91.00, Grade: Distinction
Roll No: 105, Name: Tom, Total: 93, Average: 31.00, Grade: Fail [20 Marks]

TASK 2:

Write a bash script that takes a single word as a command-line argument and counts the total number of occurrences of that word in all .csv files located in the current directory (non-recursive). The search should be:

1. Case-insensitive (e.g., "Apple" and "apple" both count)
2. Whole-word matches only (e.g., "apple" counts, but "applepie" does not)

Requirements:

1. Your script should handle a case when no .csv files exist.
2. Your script should print the total count of all occurrences across all .csv files.
3. The script should handle multiple .csv files in the current directory.

Input: A single command-line argument representing the word to search for one or more .csv files located in the current directory.

Example CSV file 1: data.csv

```
ID,Name,Fruit,Quantity
1,Alice,Apple,10
2,Bob,Banana,5
3,Charlie,apple,7
4,David,Grape,3
5,Eve,Applepie,4
6,Frank,APPLE,8
```

Sample CSV file 2: sales.csv

```
OrderID,Customer,Product,Quantity
101,John,Orange,12
102,Linda,apple,5
103,Mark,Banana,7
104,Susan,APPLE,10
105,Tom,AppleSauce,3
106,Emma,grape,8
107,Oliver,apple,6
```

Expected Output Format:

After running your script with a word to search, output a single line in this format:

Total occurrences of '<word>': <count>

For example, running ./count_word.sh apple should print:
Total occurrences of 'apple': 6 [20 Marks]